

Introduction to Deep Learning

The Team

Lecturer



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TAs



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Shaotong
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Gomes

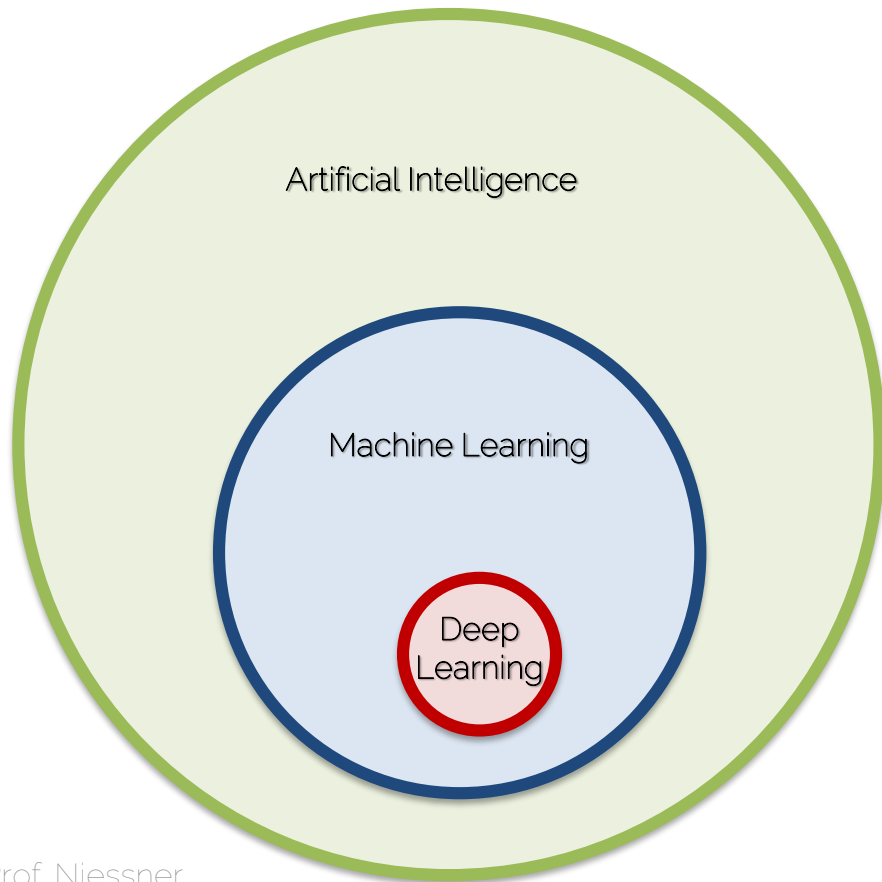


Ignacio
Dassori

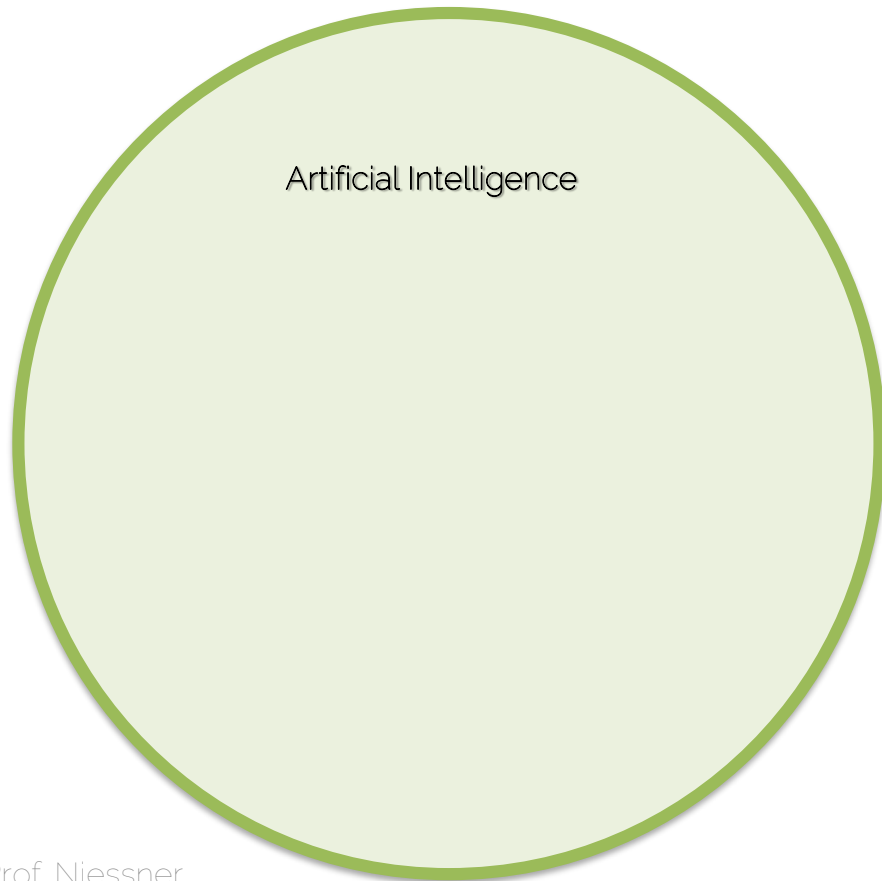


Youssef
Youssef

What is this lecture about?

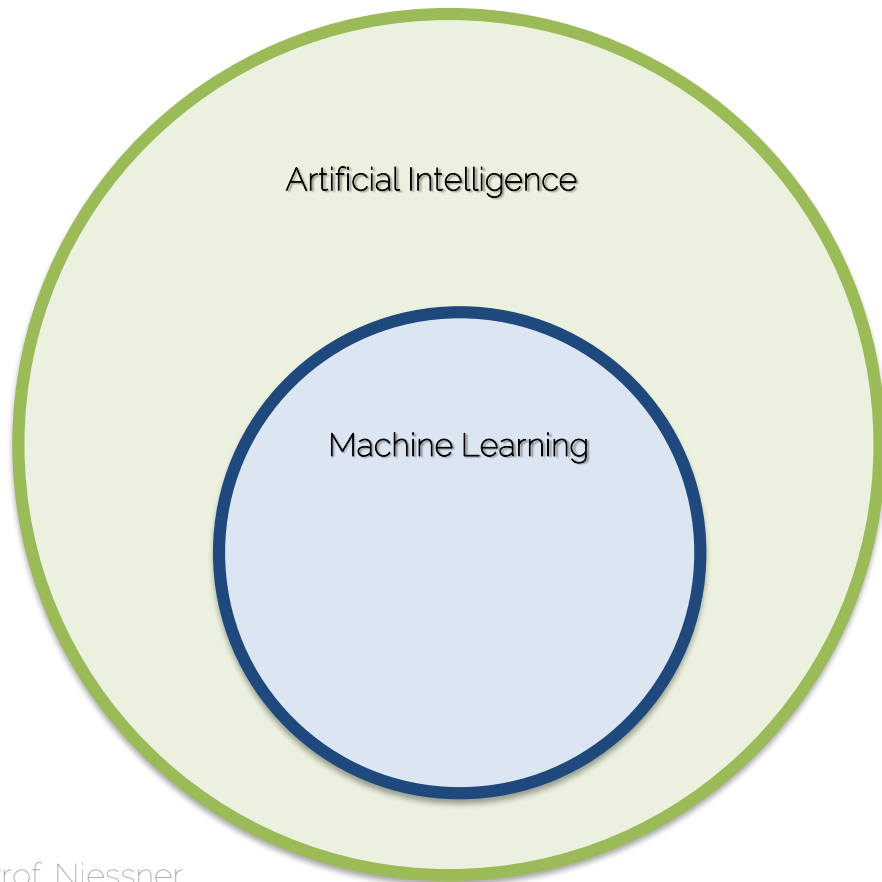


What is this lecture about?



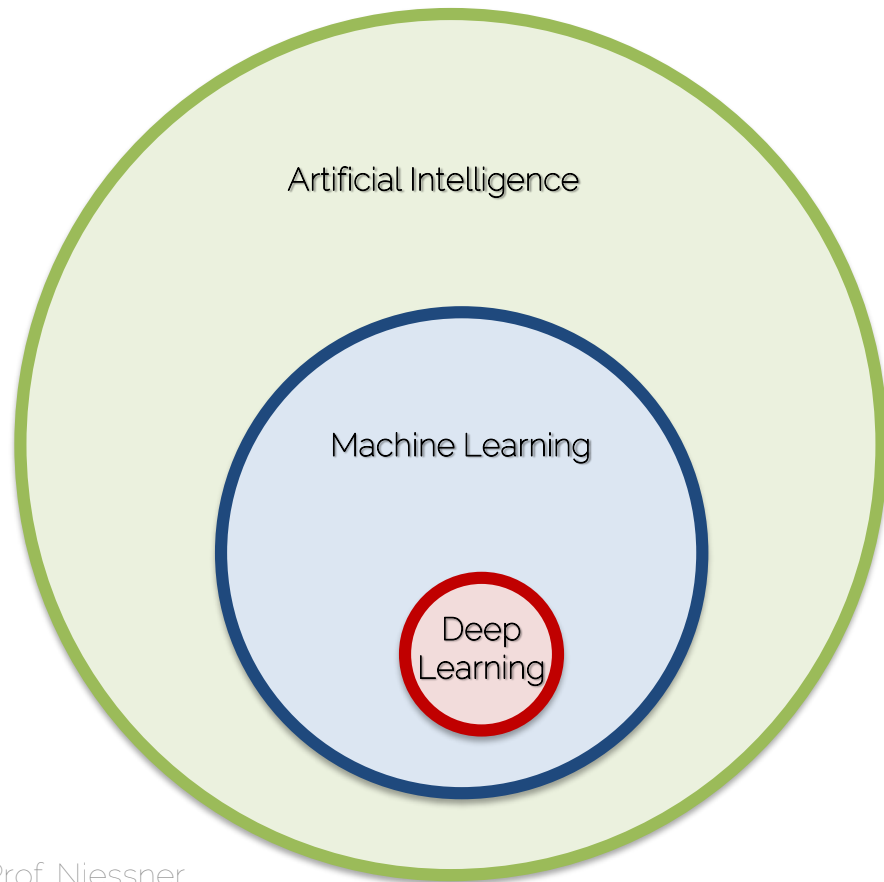
- AI Methods
 - Broad definition!
 - "if" statements"
if (cold)
turn_up_heat();
 - Binary Search
 - Dijkstra, A*, ...
 - Prime, Kurskal, ...
 - Logic algorithms, etc.
 - ...

What is this lecture about?



- ML Methods
 - Linear/logistic regression
 - Support Vector Machines
 - Random Decision Trees, Forests, Jungles, ...
 - ...

What is this lecture about?



- Deep Learning
 - ML-methods leveraging neural networks
 - Multi-layer perceptrons
 - Convolutional neural networks
 - Recurrent neural networks
 - Transformers
 - Generative models, etc...

Computer Vision

What's in the image?



What's in the image?



Q: Is the door of the truck cab open?



Q: In this image, how many eyes can you see on the animal?



Q: How many wheels can you see in the image?

What does a machine see?



What does a machine see?



What is Computer Vision?

- First defined in the 60s in artificial intelligence groups
- “Mimic the human visual system”
- Center block of robotic intelligence



Image Classification

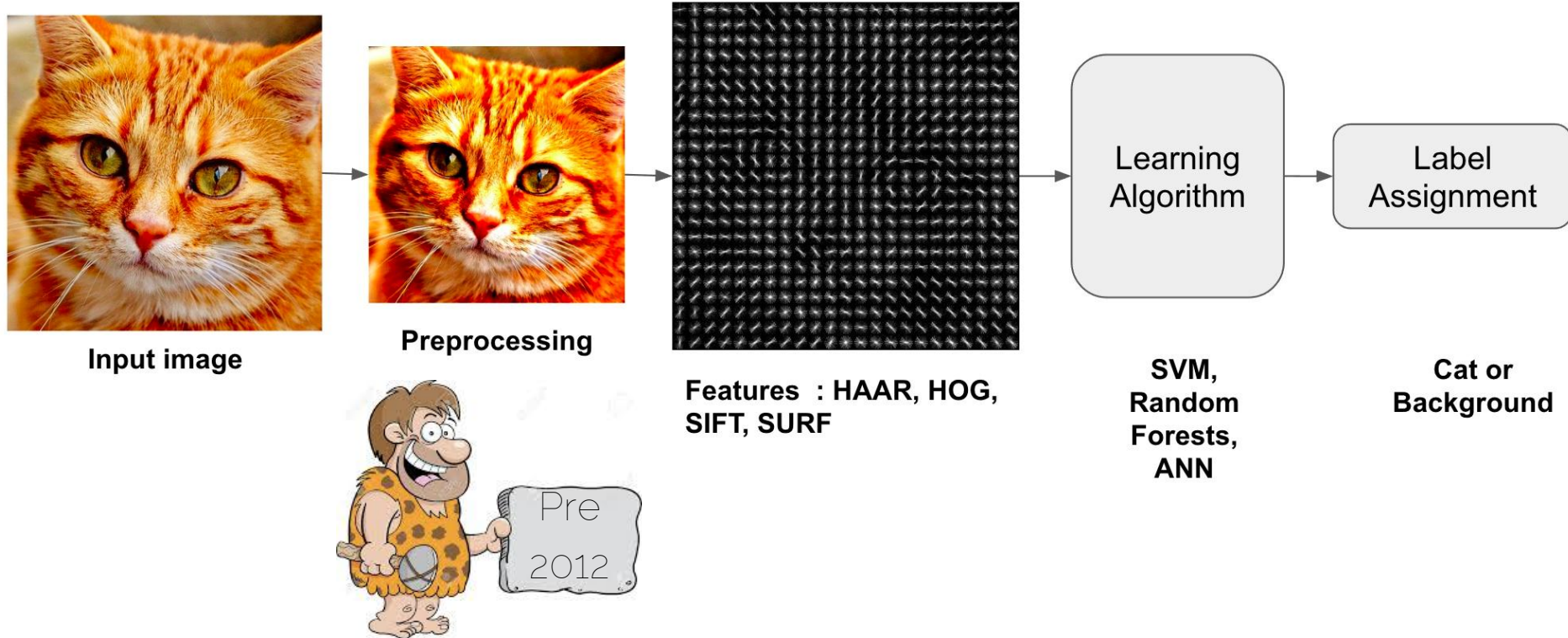


Image Classification



Input image



Deep Neural
Network

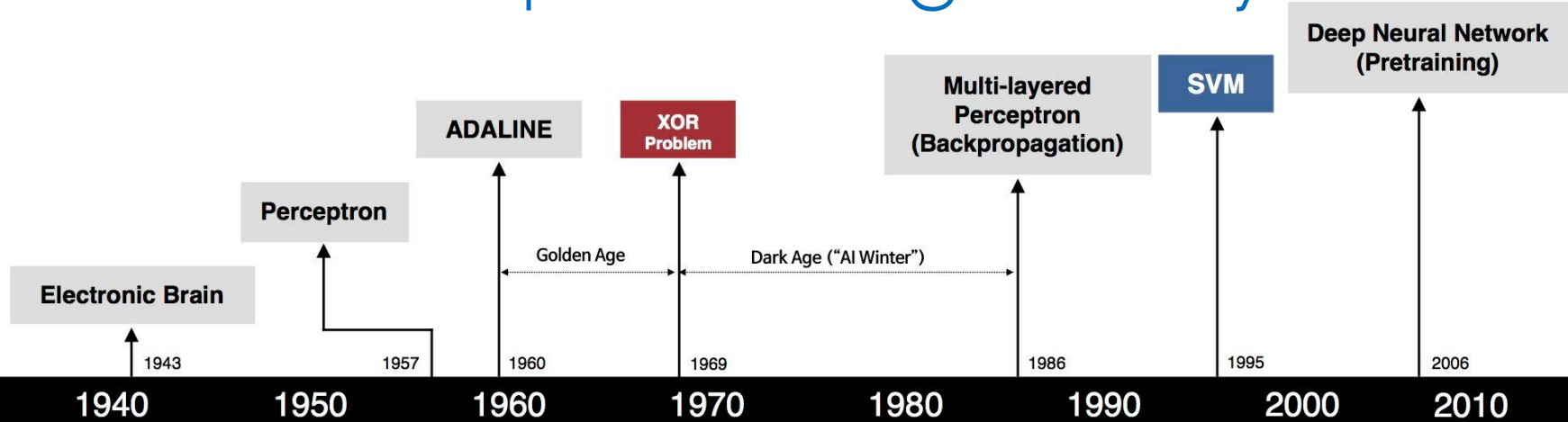


**Label
Assignment**

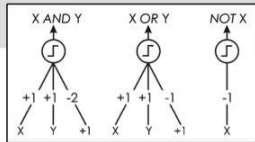
**Cat or
Background**

Why Deep Learning?

Deep Learning History



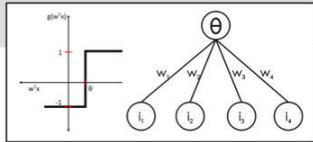
S. McCulloch – W. Pitts



- Adjustable Weights
- Weights are not Learned



F. Rosenblatt



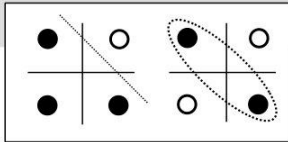
- Learnable Weights and Threshold



B. Widrow – M. Hoff



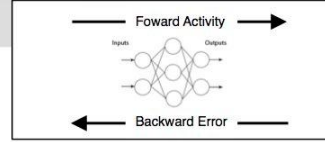
M. Minsky – S. Papert



- XOR Problem



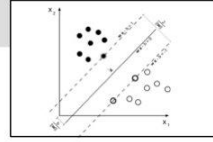
D. Rumelhart – G. Hinton – R. Williams



- Solution to nonlinearly separable problems
- Big computation, local optima and overfitting



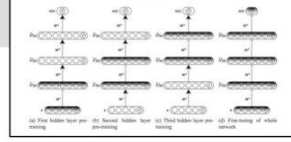
V. Vapnik – C. Cortes



- Limitations of learning prior knowledge
- Kernel function: Human Intervention

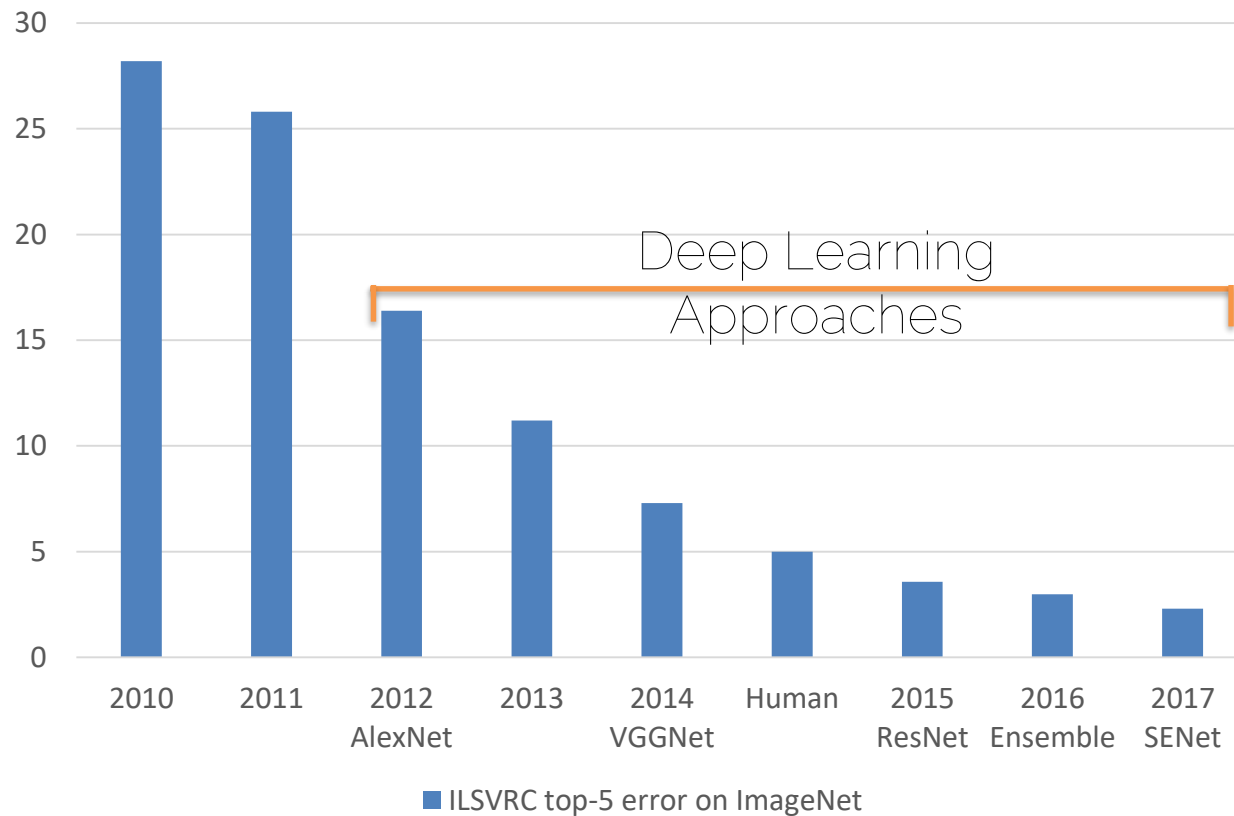


G. Hinton – S. Ruslan



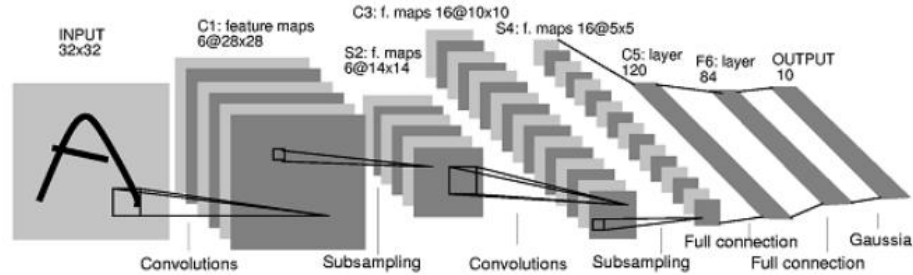
- Hierarchical feature Learning

Modern Deep Learning in Computer Vision



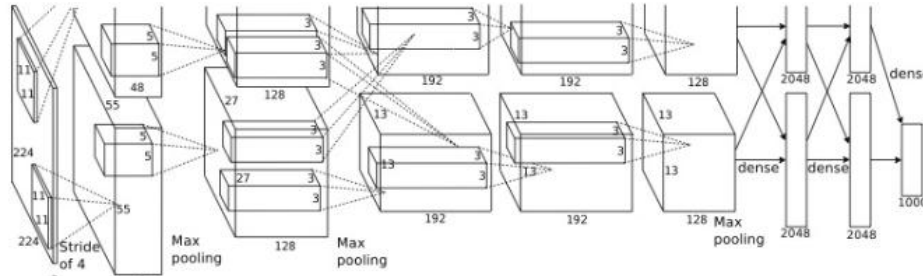
What has Changed?

1998
LeCun
et al.



- MNIST digit recognition dataset
- 10^7 pixels used in training

2012
Krizhevsky
et al.



- ImageNet image recognition dataset
- 10^{14} pixels used in training

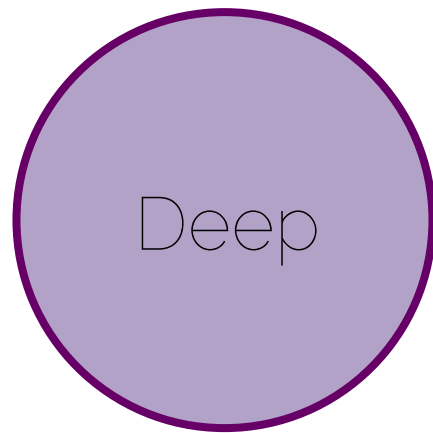
What Made this Possible?



Models know
where to learn from

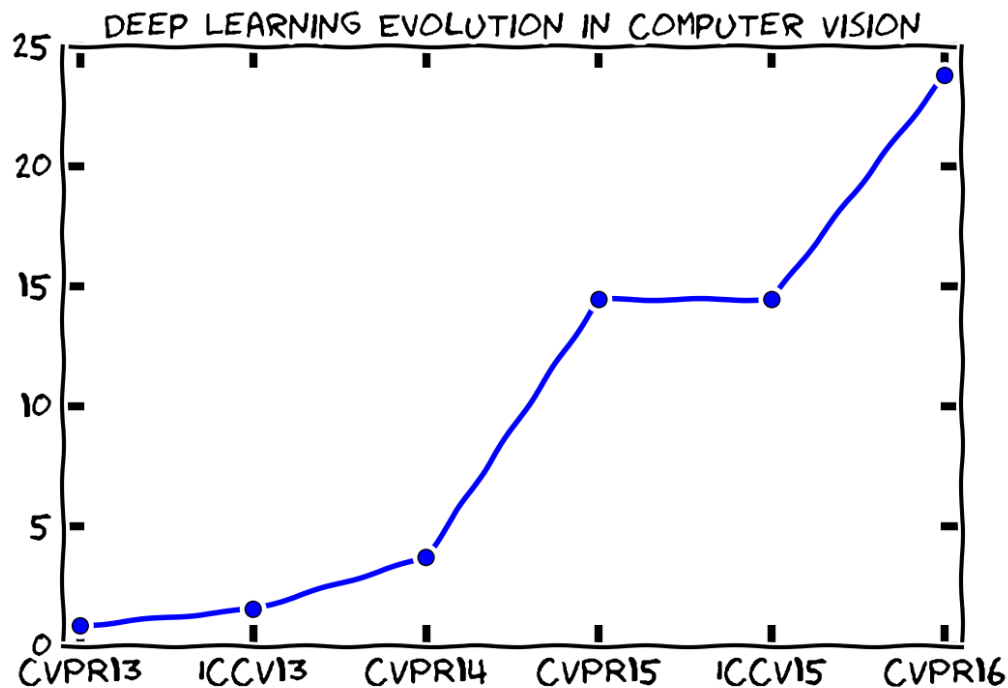


Models are
trainable



Models are
complex

Deep Learning and Computer Vision



There are now >2,000 papers at almost every top-tier AI conference (CVPR/ICCV/ECCV NeurIPS/ICLR/ICML)

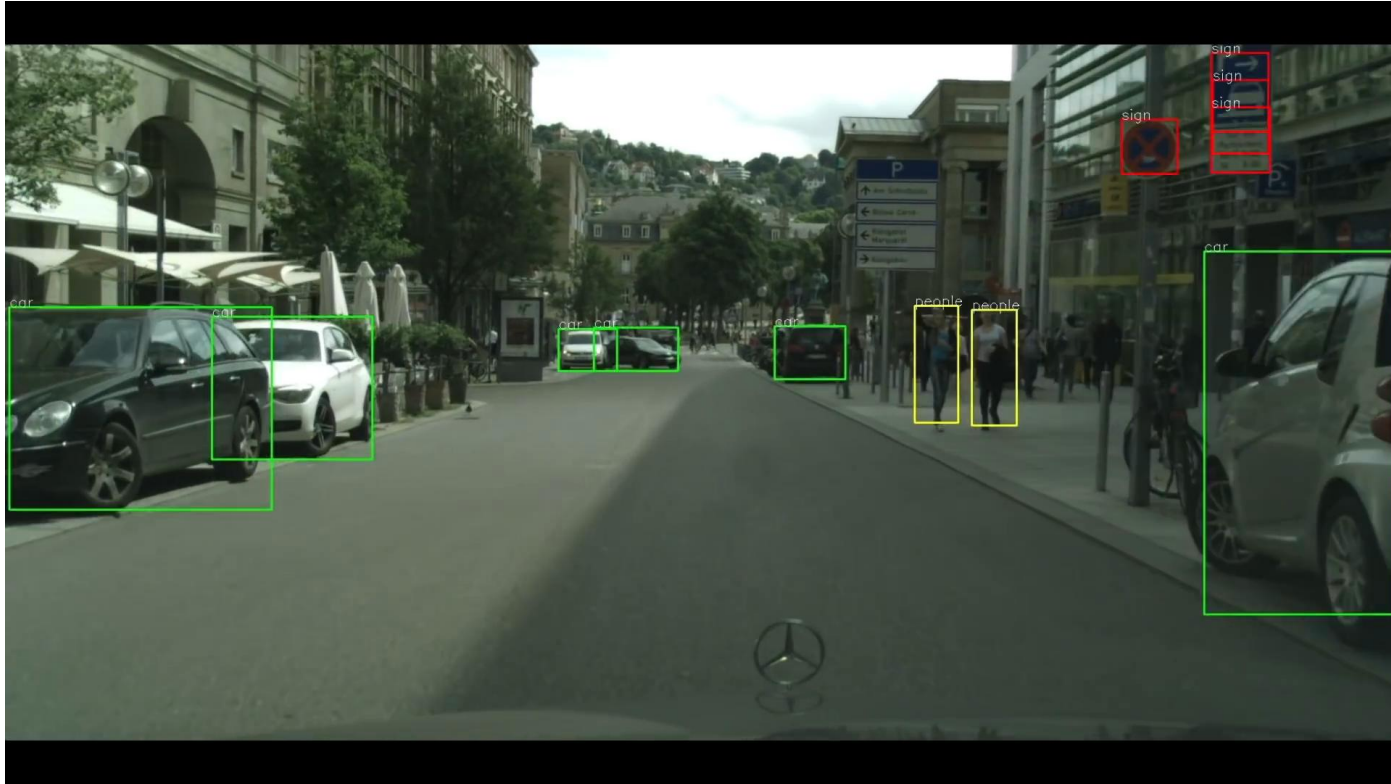
Credits: Dr. Pont-Tuset, ETH Zurich

Deep Learning Today



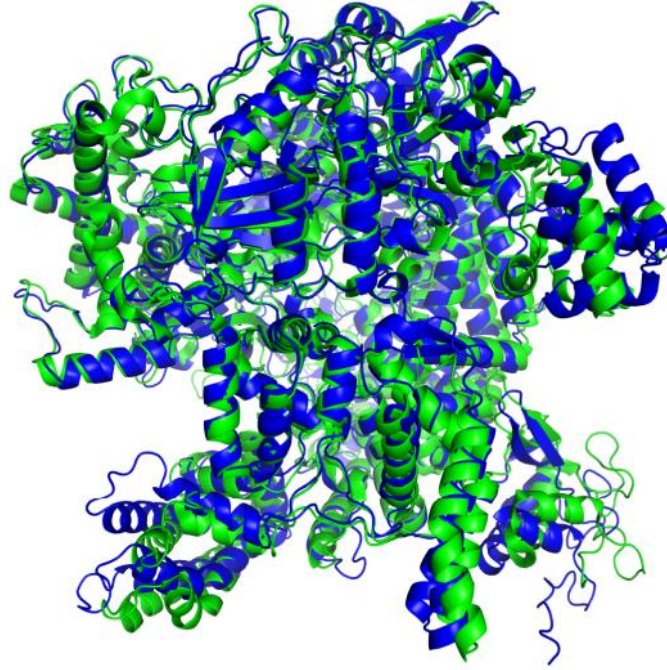
Object Detection

Deep Learning Today



Self-driving cars

Deep Learning Today



AlphaFold Experiment

Biological research

Deep Learning Today

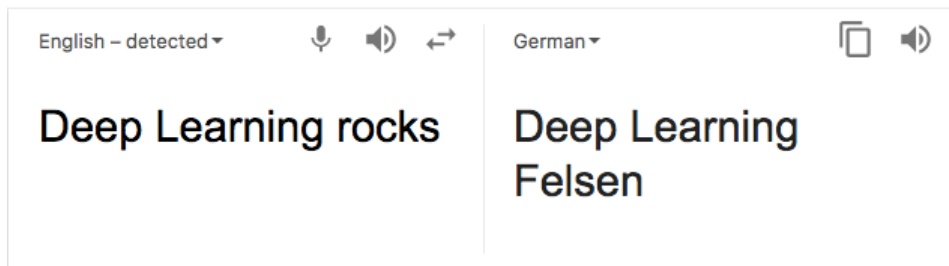


AlphaGo

ever punch a cactus?



Emoticon suggestion



Machine translation

The figure illustrates the MaNa framework for StarCraft II. At the top, two game interface screenshots are shown: the left one is labeled 'AlphaStar' and 'Render of Agent's view', showing a top-down view of the game; the right one is labeled 'MaNa' and shows a different perspective. Below these, a flow diagram shows 'Raw Observations' (a map) being processed by 'Neural Network Activations' (three heatmaps) to produce 'Considered Location' (a heatmap) and 'Considered Build/Train' (a list of units). An 'Outcome Prediction' graph shows the probability of 'Win', 'Draw', or 'Lose' over time.

Deep Learning Today



Google LaMDA (Google IO'22)

Deep Learning Today

- Large Language Models (LLMs), e.g., Chat-GPT



Tell me about deep learning and how I can teach it at TUM.



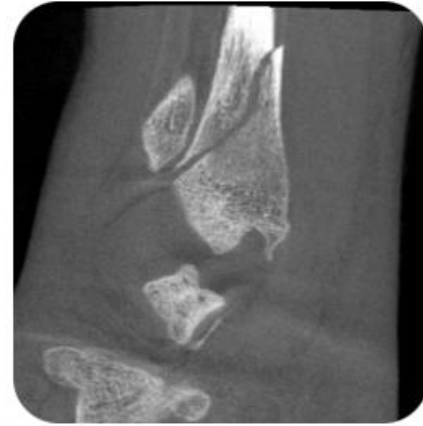
Deep learning is a subset of machine learning that involves training artificial neural networks to recognize patterns and make predictions or decisions based on input data. The term "deep" refers to the fact that these networks have many layers, which allows them to learn increasingly complex representations of the data as it moves through the network.



To teach deep learning at TUM, you could start with an introductory course that covers the basics of machine learning and neural networks. This could include topics such as:

1. Supervised learning: This involves training a model using labeled examples, so that it can make predictions on new, unseen data.

Deep Learning Today



This is a scan of my ankle, showing the fibula. Anything wrong? Short answer only; need additional assessment to radiologist.

Short answer: **Yes, this looks abnormal.**

There appears to be a **fracture of the distal fibula** with **displacement of bone fragments**, which can indicate an unstable injury.

Deep Learning Today

TEXT DESCRIPTION

An astronaut Teddy bears A bowl of
soup

mixing sparkling chemicals as mad
scientists shopping for groceries working
on new AI research

in the style of ukiyo-e as a one-line
drawing in ancient Egypt



DALL-E 2



[Dall-E 2](#) Image generation from text

Deep Learning Today

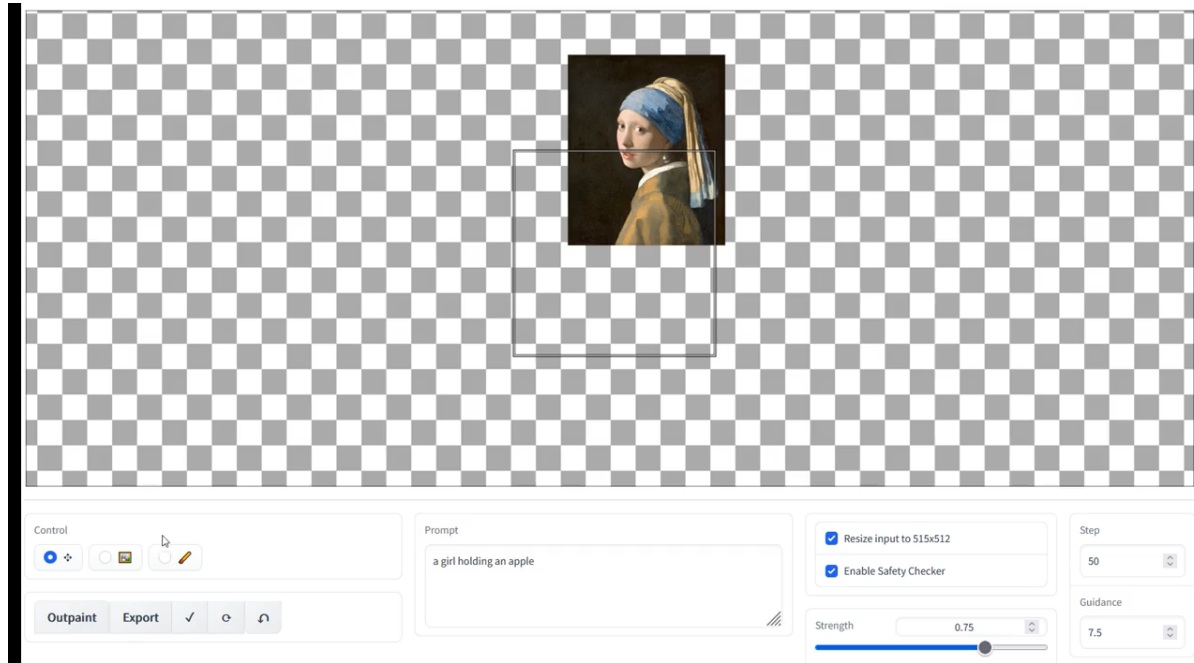


"a robot painting a colorful sunset on a canvas in a futuristic art studio"



"a majestic dragon made of water flying through a colorful, abstract city"

Deep Learning Today



StableDiffusion Image Outpainting

Deep Learning Today



Deep Learning Today



Deep Learning Today



Deep Learning Today



Deep Learning Today



Deep Learning Today



Deep Learning Today



Deep Learning Today



Image to 3D World

Deep Learning Today

The screenshot shows a VS Code editor interface with a file explorer on the left, a code editor in the center, and a chat window on the right. The code editor displays a Python class `CoordMLP` with methods for building a neural network. The chat window contains a question about dead ReLUs and a detailed answer explaining the approach and providing a code snippet.

```
class CoordMLP(nn.Module):
    def __init__(self, activation, skip_layer=None, num_layers=5, hidden_dim=60, out_features=1000):
        super().__init__()
        self.activation = activation
        self.skip_layer = skip_layer
        self.num_layers = num_layers
        self.hidden_dim = hidden_dim
        self.out_features = out_features

        act_cls = self.ACTIVATIONS[activation]
        pre = list(nn.Module) = []
        prev_dim = encoded_dim
        for _ in range(skip_layer):
            pre.append(nn.Linear(prev_dim, hidden_dim))
            pre.append(act_cls())
            prev_dim = hidden_dim
        self.net_pre = nn.Sequential(*pre)

        post = list(nn.Module) = []
        prev_dim = hidden_dim + encoded_dim # skip concat
        for _ in range(skip_layer, num_layers):
            post.append(nn.Linear(prev_dim, hidden_dim))
            post.append(act_cls())
            prev_dim = hidden_dim
        post.append(nn.Linear(prev_dim, out_features))
        post.append(nn.Sigmoid())
        self.net_post = nn.Sequential(*post)

    else:
        act_cls = self.ACTIVATIONS[activation]
        layers = []
        prev_dim = encoded_dim
```

is there any easy way to figure out how many dead ReLUs are actually in that training run?

Approach: For each ReLU layer, track whether each neuron ever produces a positive output across all input pixels. Any neuron that stays at zero for every single input is "dead."

Since you have the saved checkpoints, you'd do something like this:

```
import torch
from model import CoordMLP
from dataset import ImageDataset

device = torch.device('cuda')
dataset = ImageDataset('data/milkyway.png')
coords = dataset.coords.to(device)

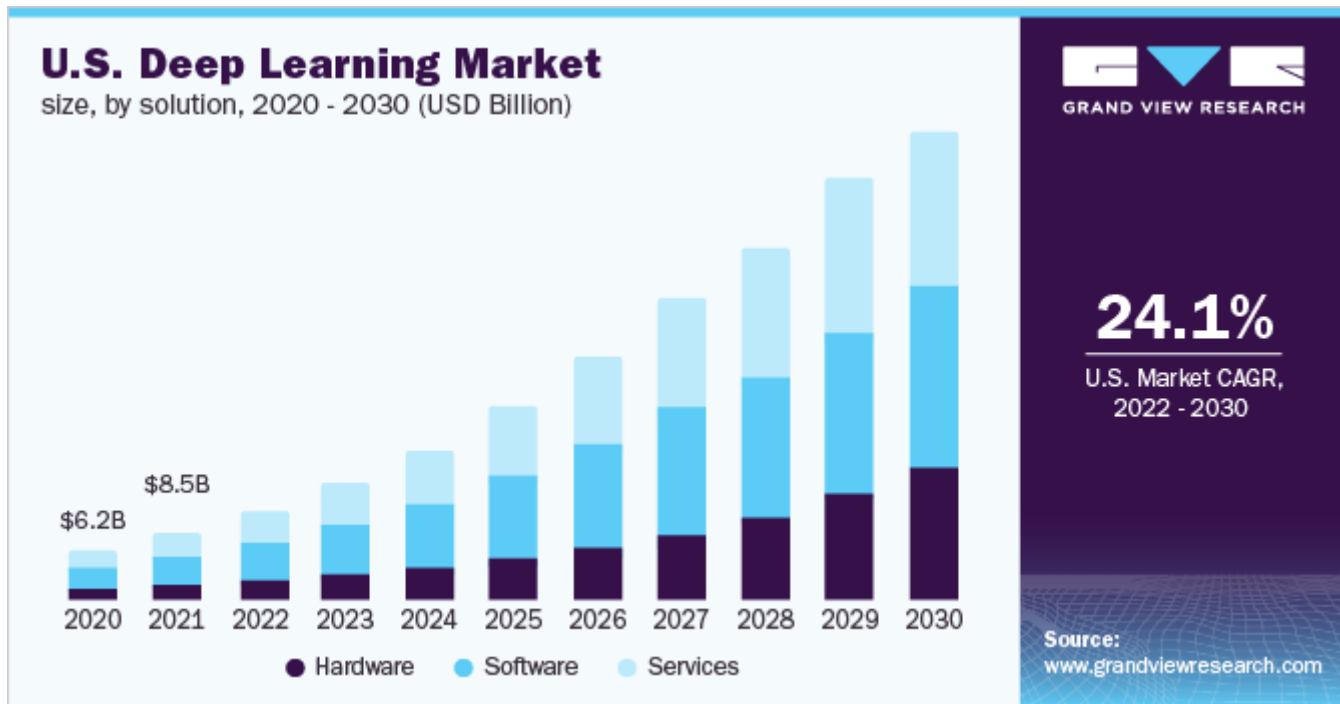
for label, path in [(('from_scratch', 'checkpoints/milkyway_from_scratch.pth'),
                    ('from_blackhole', 'checkpoints/milkyway_from_blackhole.pth'))]:
    model = CoordMLP(activation='relu', hidden_dim=60, num_layers=5,
                    encoding='positional', skip_layer=2).to(device)
    model.load_state_dict(torch.load(path, map_location=device, weights_only=True))
    model.eval()

    # Track which neurons ever activate (per ReLU layer)
    ever_active = {}
    hooks = []
    for name, module in model.named_modules():
        if isinstance(module, torch.nn.ReLU):
            def make_hook(n):
                def hook_fn(mod, inp, out):
                    mask = (out > 0).any(dim=0) # any pixel activated this neuron?
                    if n not in ever_active:
                        ever_active[n] = mask
            hooks.append(module.register_forward_hook(make_hook(n)))

    model.eval()
    torch.no_grad()
    outputs = model(coords)
    for hook in hooks:
        hook.remove()

    dead_relus = sum(1 for n, mask in ever_active.items() if not mask.any())
    print(f"Dead ReLUs in {name}: {dead_relus}")
```

Deep Learning Market (estimated 2021)



[...] market research report Deep Learning Market [...] " the deep learning market is expected to be worth **USD 415 Billion by 2030.**

Deep Learning Job Perspective

- Excellent Job Perspectives!
 - Automation requires ML/DL -> growth!
 - Top-notch companies will gladly hire you!
- Many industries now:
 - IT-Companies
 - Cars, Logistics, Health Care, etc...
 - Manufacturing / Robotics, etc...

But also Challenging!

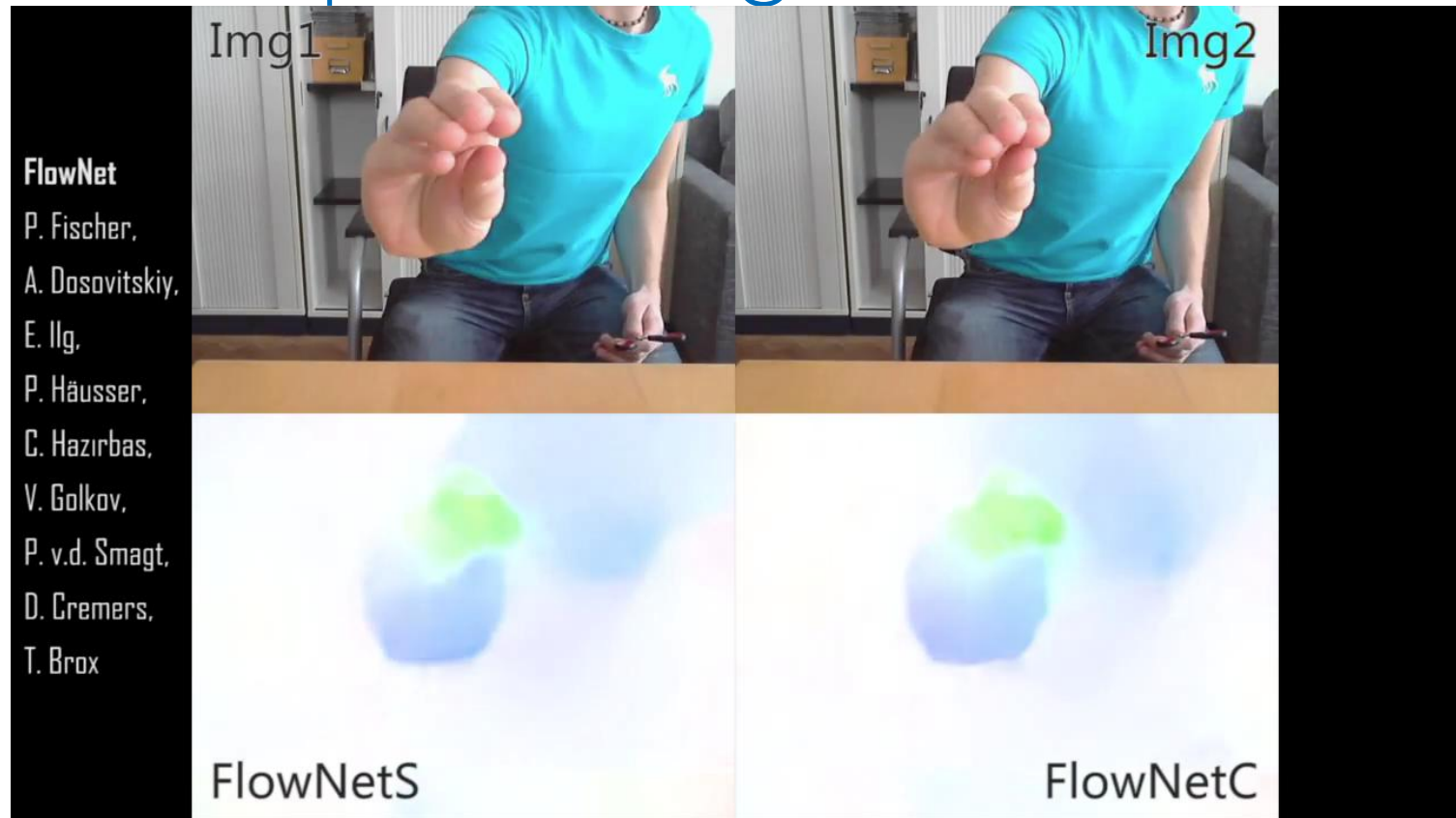
- High-level understanding is not enough
 - Lots of consumers, few creators
 - Need proper theory background
 - Need proper practical skillsets
- Can be competitive!
 - Many good people
 - Education gap widens
 - Running scripts / implementation also by AI 😊
 - Deeper understanding mostly requires a PhD

Deep Learning at TUM

Many TUM Research Labs use DL

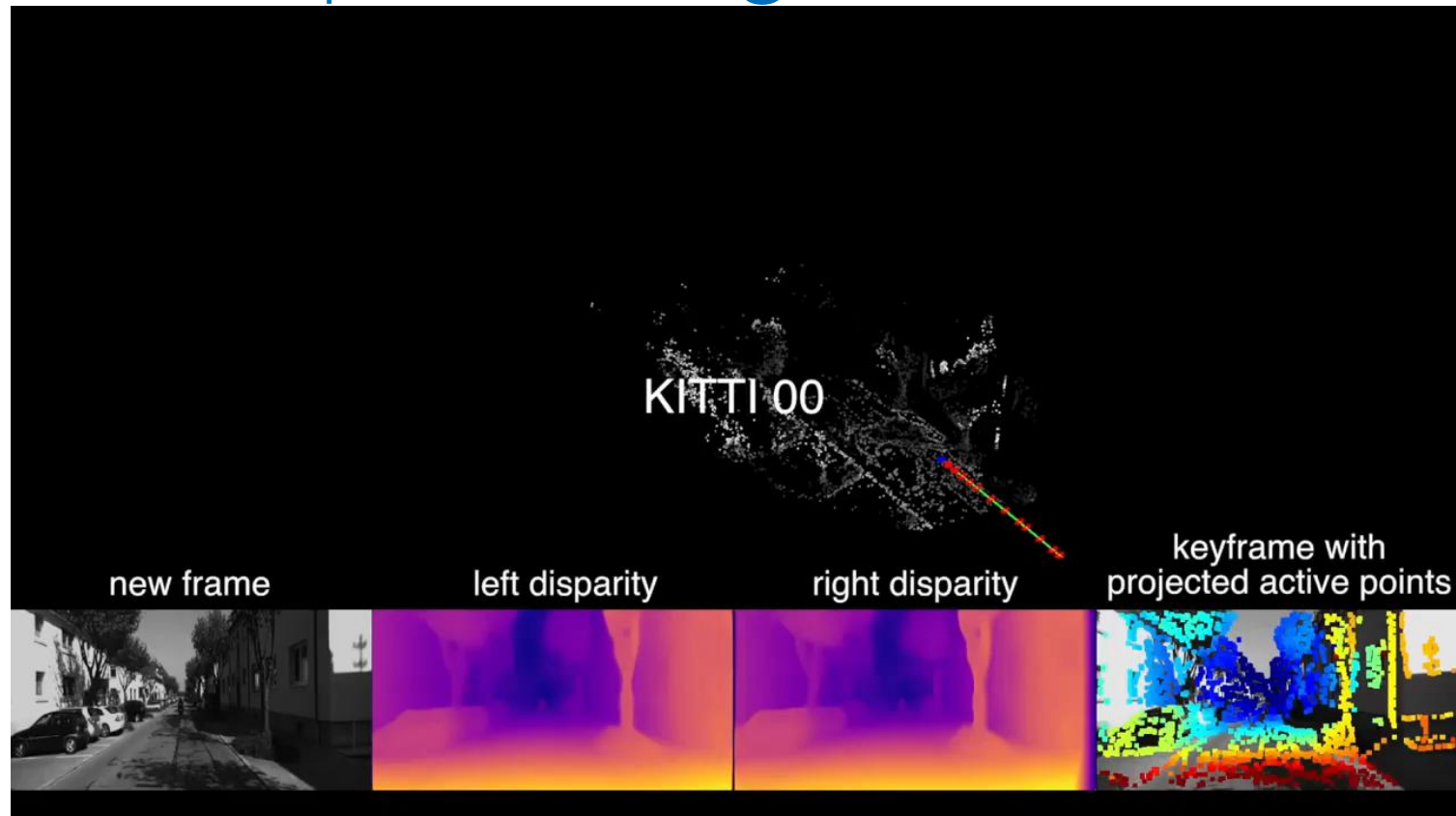
- 3D AI Lab (Prof. Dai)
 - Research in 3D perception, 3D scene understanding
- Visual Computing & AI (Prof. Niessner):
 - Research in computer vision, graphics, and machine learning
- Interpretable and Reliable Machine Learning (Prof. Akata-Schulz)
 - Research in NLP and medical domains
- Robotics and Artificial Intelligence (Prof. Schoellig)
 - Research in Robotics and machine learning
- Computer Vision Group (Prof. Cremers)
 - Research in computer vision, machine learning and robotics
- Data Mining and Analytics Lab (Prof. Günnemann)
 - Research methods for robust machine learning
- Computer Aided Medical Procedures (Prof. Navab)
 - Research in machine learning for medical applications
- And many more 😊

Deep Learning at TUM



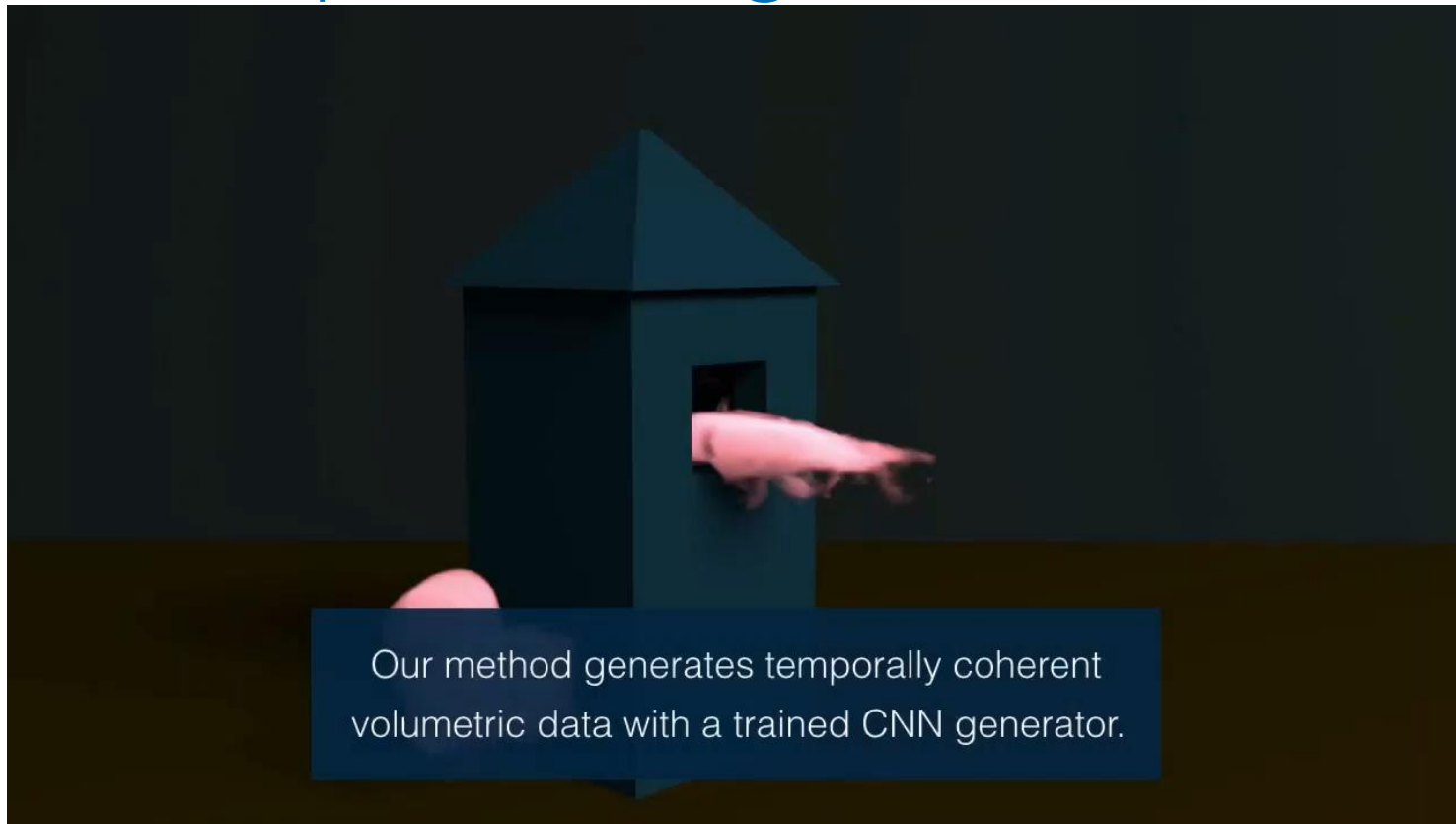
[Dosovitskiy et al., ICCV' 15] FlowNet

Deep Learning at TUM



[Yang et al., ECCV'18] Deep Virtual Stereo Odometry

Deep Learning at TUM



[Xie et al. Siggraph' 18] tempoGAN

Deep Learning at TUM

Animation Synthesis



[Thies et al., Siggraph'19]: Neural Textures

Deep Learning at TUM

Animation Synthesis

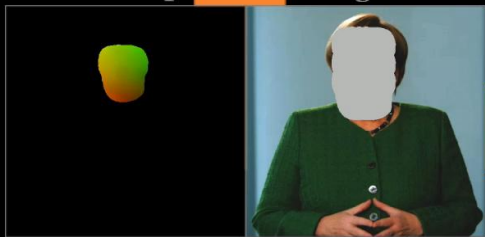
Source Actor



Target
UV-Map



Target
Background



Output



[Thies et al., Siggraph'19]: Neural Textures

Deep Learning at TUM



[Kirchstein et al., ToG'23] Nersemble

Deep Learning at TUM



ScanNet Stats:

- Kinect-style RGB-D sensors
- 1513 scans of 3D environments
- 2.5 Mio RGB-D frames
- Dense 3D, crowd-source MTurk labels
- Annotations projected to 2D frames

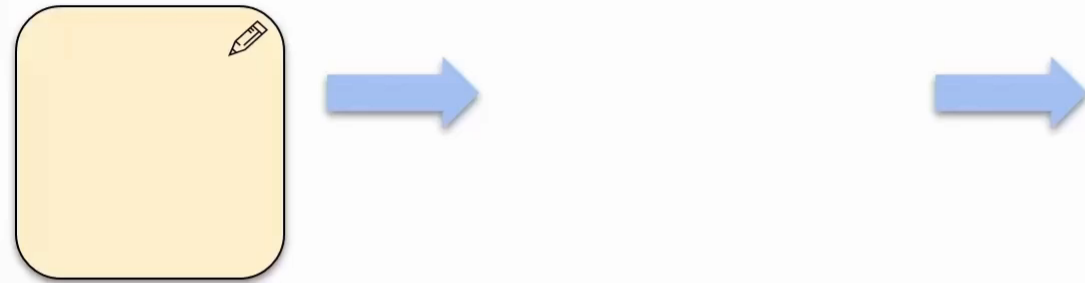
Deep Learning at TUM



- ScanNet++:
- 1006 high-fidelity Scenes
 - 1mm laser scans
 - DSLR images
 - iPhone RGB-D
 - semantics

[Yeshwanth & Liu et al., ICCV'23] ScanNet++

Deep Learning at TUM



Deep Learning at TUM



3D Semantic Map

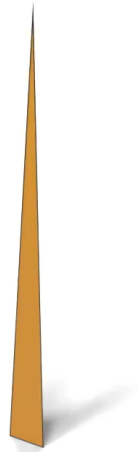


Scene Geometry



[Bokhovkin et al., CVPR'25] SceneFactor

Deep Learning at TUM



[Siddiqui et al., CVPR'24] MeshGPT

Deep Learning at TUM



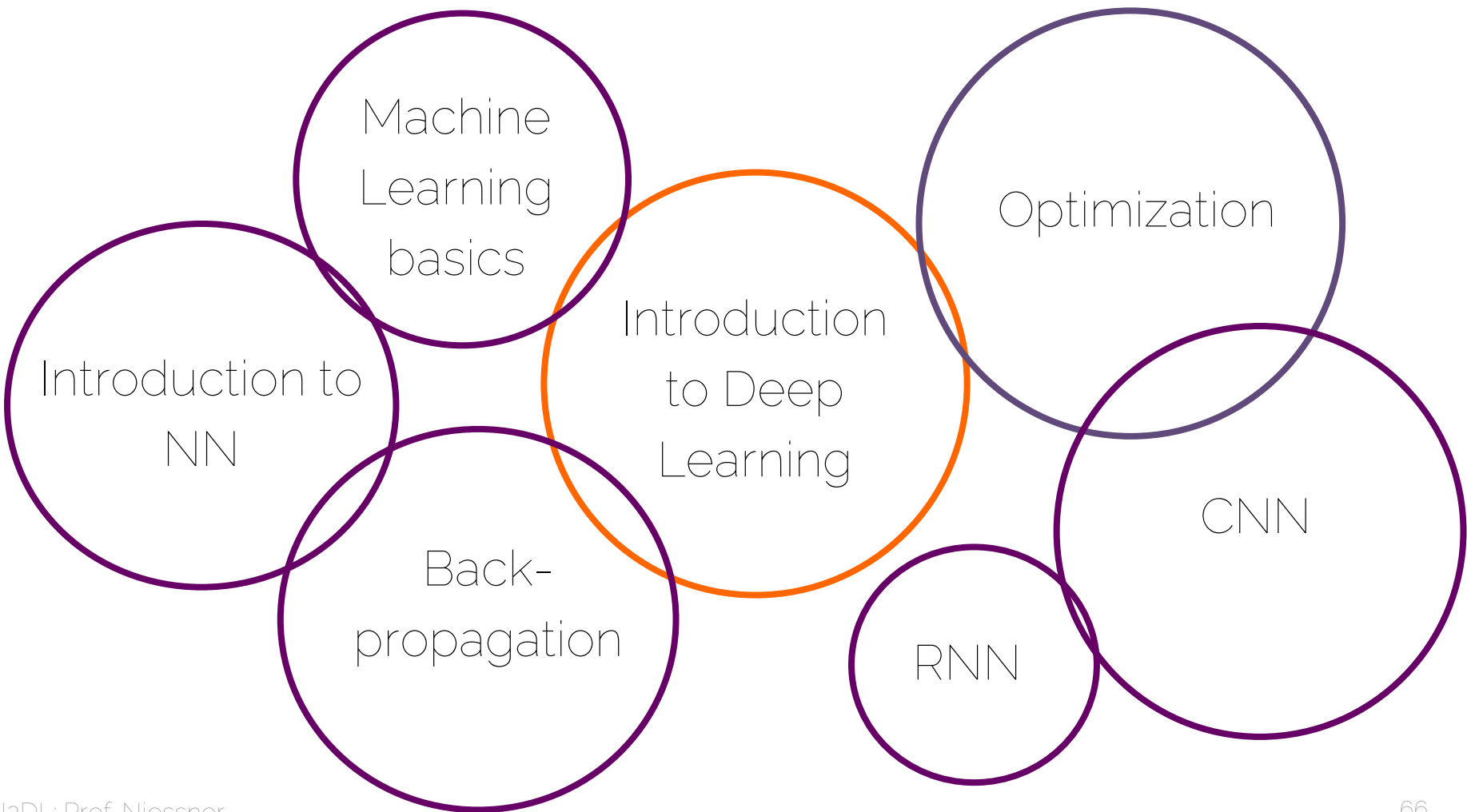
[Schneider et al., SIGGRAPH Asia'25] WorldExplorer

Deep Learning at TUM

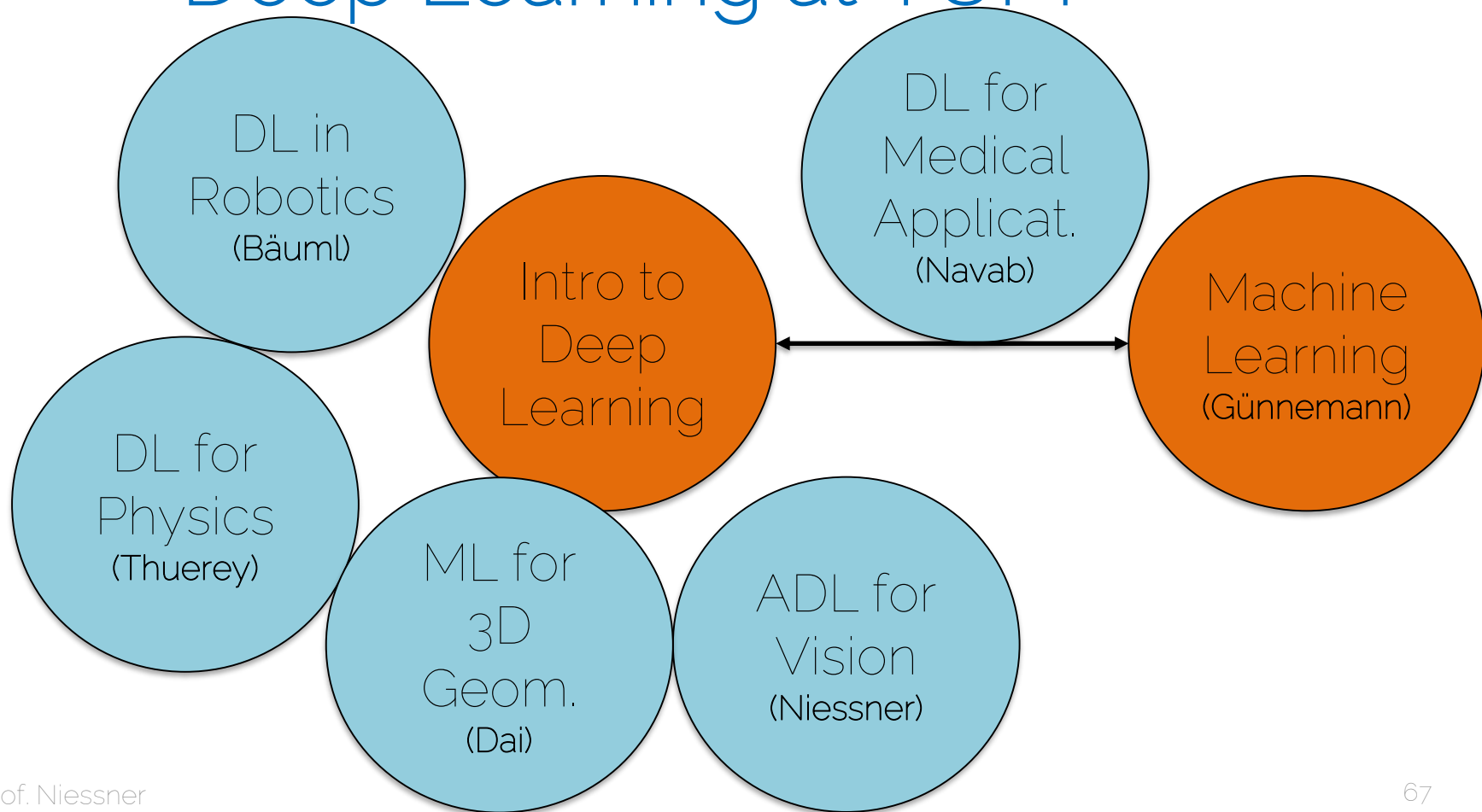


[von Lützwow et al., arXiv'26] GaussianGPT

Relation to other lectures at TUM



Deep Learning at TUM



Deep Learning at TUM

- I2DL is critical for the curriculum:
 - often a pre-requisite for many other lectures
 - many lectures build on top of it
- Follow up lectures such as Advanced Deep Learning for Visual Computing (generative models etc.)
- Take I2DL first -> AdvancedDL -> practical/GR/IDP

Introduction to Deep Learning

Logistics

About the Lecture

- Course webpage: <https://niessner.github.io/l2DL/>
- Theory lectures (every Monday at 14:15)
 - In-person, live-streamed
- Tutorials and exercises (every Thursday at 10:00)
 - Tutorial: Online videos posted to Moodle and the webpage
 - Practical exercises
- Guest Lecture 😊

Preliminary Syllabus

- Lecture 1: Introduction to the lecture, Deep Learning, Machine Learning.
- Lecture 2: Machine Learning Basics, Linear regression, Maximum Likelihood
- Lecture 3: Introduction to Neural Networks, Computational Graphs
- Lecture 4: Optimization and Backpropagation
- Lecture 5: Scaling Optimization to large Data, Stochastic Gradient Descent
- Lecture 6: Training Neural Networks I
- Lecture 7: Training Neural Networks II
- Lecture 8: Training Neural Networks III
- Lecture 9: Introduction to CNNs
- Lecture 10: CNNs architectures
- Lecture 11: Recurrent Neural Networks (RNNs)
- Lecture 12: Transformers

Moodle

- Announcements via Moodle - **IMPORTANT!**
 - We will share common information (e.g., regarding exam)
- The Q&A Forum
 - Ask and discuss questions
 - Tutors will monitor and answer questions
 - You are very welcome to actively participate
 - Please do not post solutions of the exercises
- Private Questions: email i2dl@vc.in.tum.de

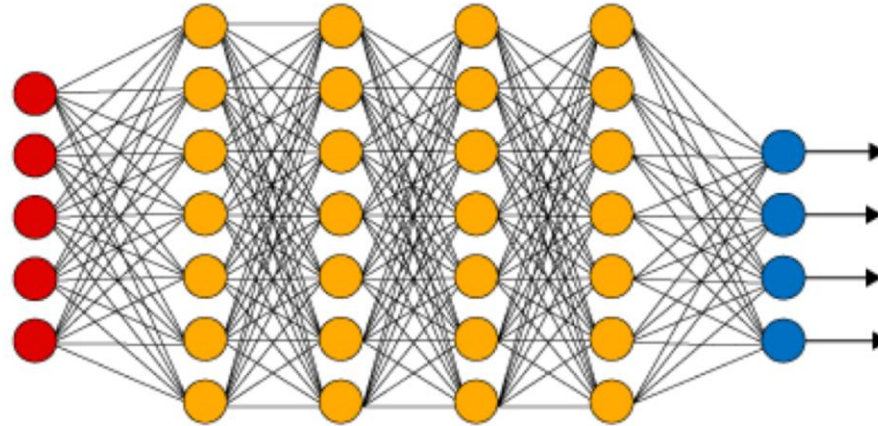
Email

- Email list:

i2dl@vc.in.tum.de

- Do NOT email us personally!
 - Cannot handle so many emails / hence will be ignored
- Email list for organizational questions only!
 - Content questions -> Moodle or Office Hours

Introduction to Deep Learning (I2DL) (IN2346)



<https://niessner.github.io/I2DL/>

(Virtual) Office Hours

- We will have dedicated office hours regarding
 - Theoretical help (e.g., specific lecture questions)
 - Help on exercises
- More info in the first tutorial session
- Video links will be posted on Moodle

Exam FAQ

- Final Exam: 29.07, Wed, 8:30-10:00
- Content: Lecture & exercises
- Important: No retake exam (I2DL is taught every semester)
- Grade Bonus:
 - Solve 8 out of 9 "non-optional" programming exercises
 - Bonus 0.3 on a **passed** final exam
 - Bonus is transferable from previous and future semesters

Other Administrative

- “External” students welcome (LMU, TUM PhD)
 - You should naturally have the Moodle access.
 - If not, fill out registration form and we will add you to the course
 - Will get Certificate / Schein at the end
- Again:
 - Check announcements on Moodle
 - Check content on website:
<https://niessner.github.io/I2DL/>

Practical Exercises

Exercise – Goal

- Hands-on programming experience (learning by doing)
- Reimplement basic building blocks
- Introduction to common libraries
- (Get grade bonus)
- Ultimately: Gather enough experience to start your own individual (research) deep learning project

Exercise – Format

- Tutorial:
 - Video only
Posted on Moodle and course website
 - Video length
Ex02: Full lecture with written exercise
Ex03-12: Short (~30min) video and coding exercises
- Programming:
 - Interactive coding notebooks (~4h each)
- Start time: Thursdays 10:00
- Working Time: 1-2 weeks
- Deadline: Wednesdays,
23:59

Upcoming Lecture

- Important:
 - Next lecture is on 20.04
- Next Lecture: Lecture 2: Machine Learning basics
- Thursday: Tutorial 1 and Exercise 1

See you next time 😊